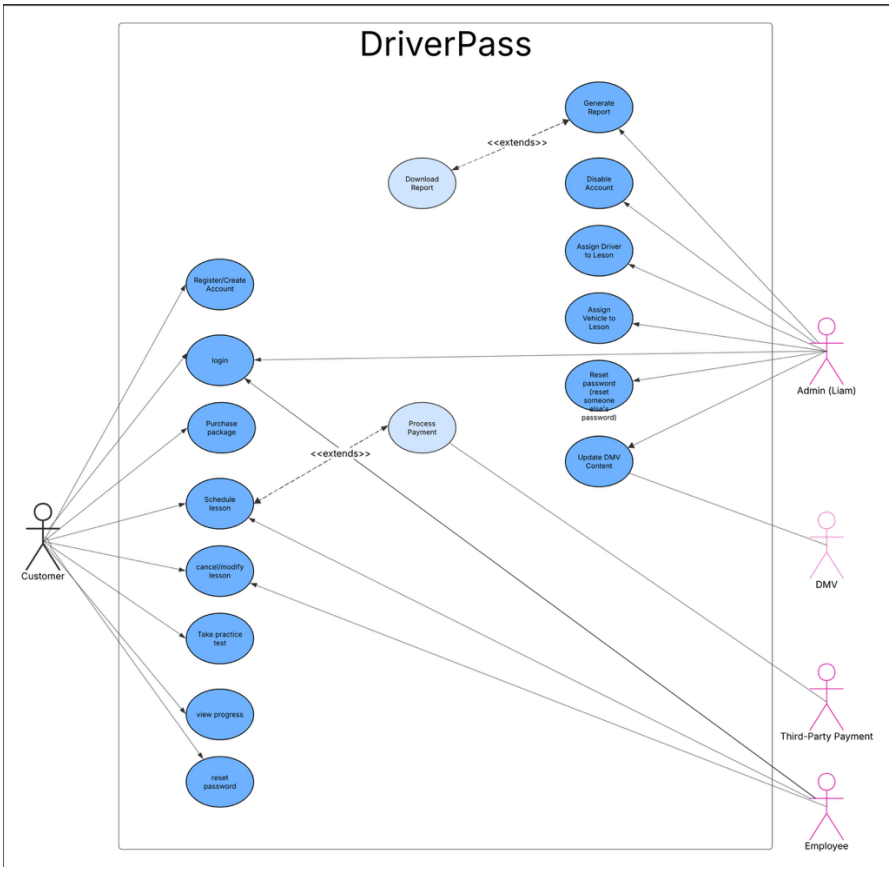
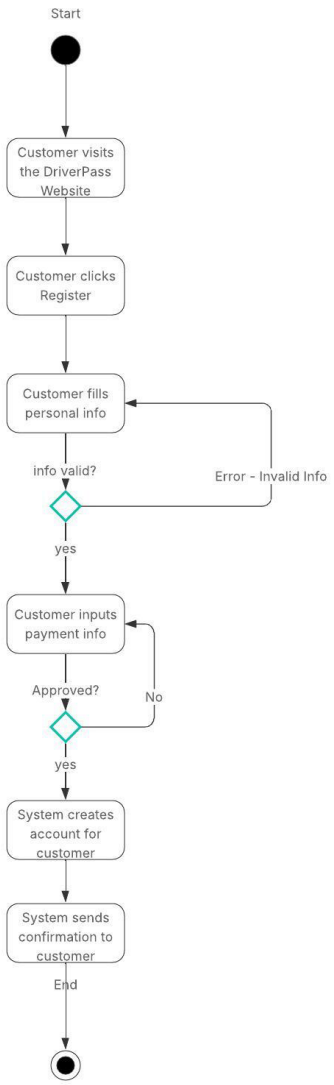
CS 255 System Design Document Template

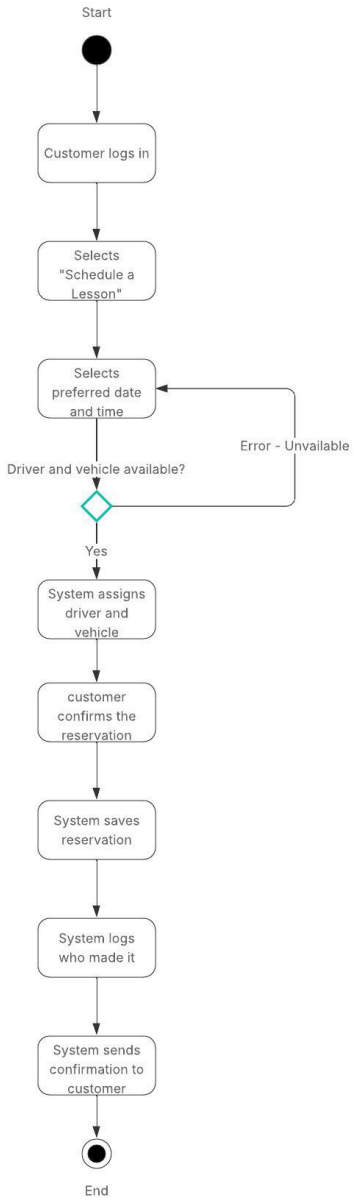
UML Diagrams

UML Use Case Diagram

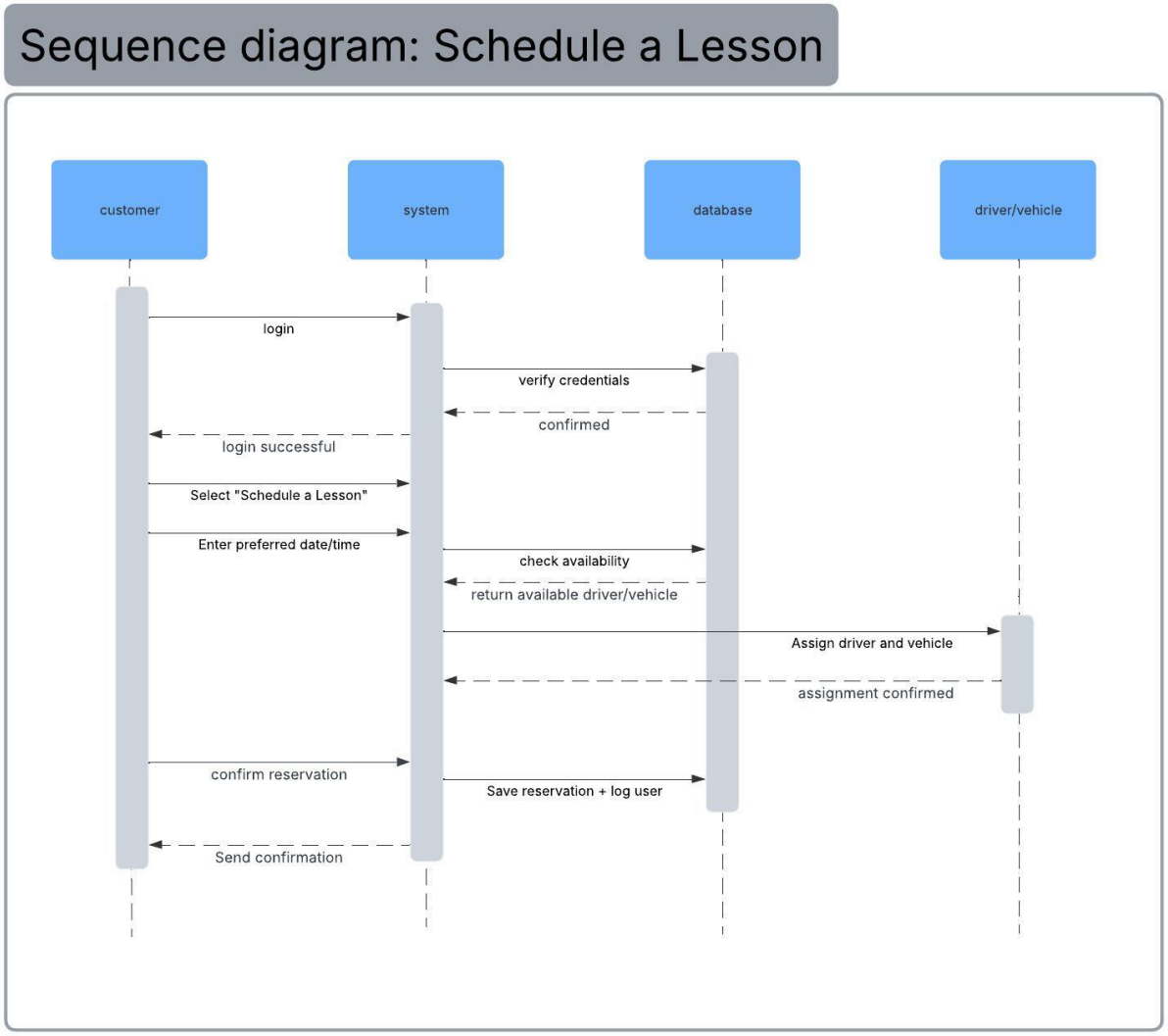


UML Activity Diagrams

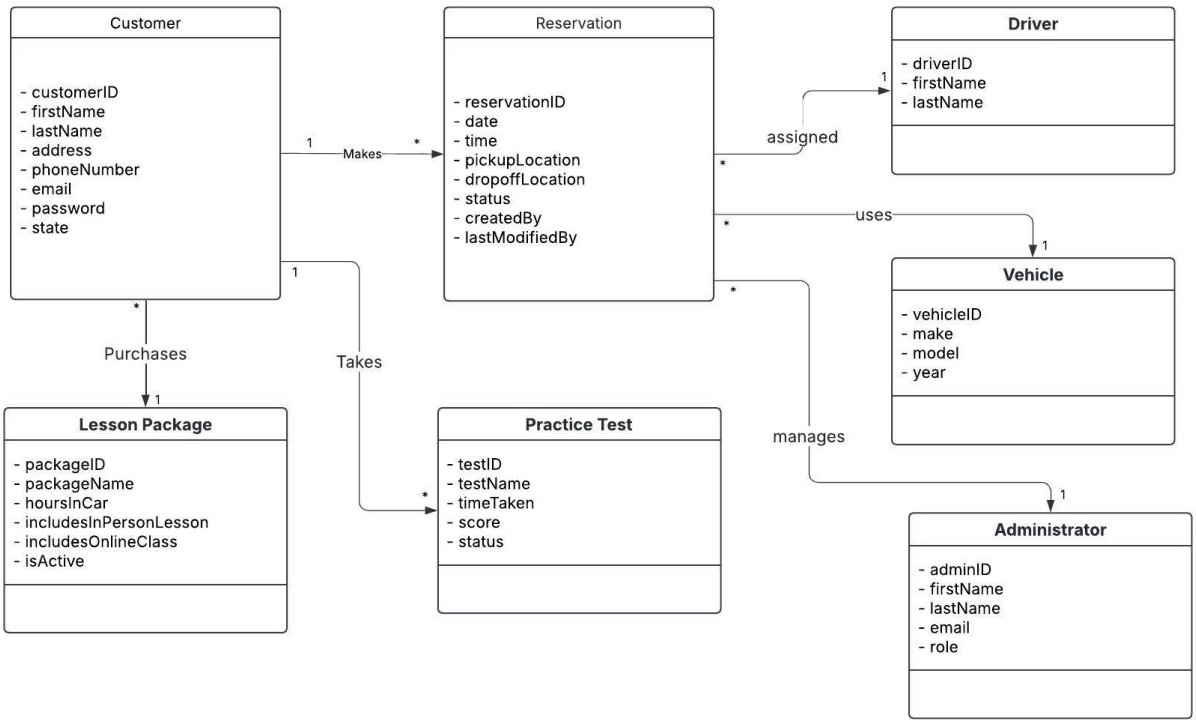




UML Sequence Diagram



UML Class Diagram



Technical Requirements

Hardware

DriverPass doesn't need any special servers or equipment because everything is hosted in the cloud. Users just need a computer, laptop, tablet, or smartphone with internet access and an updated web browser. In the DriverPass office, we'll also need a printer in case we want to print activity reports or reservation details.

Software

The system will be built as a web application using the PERN stack. The front end will use React to create a responsive and interactive user interface that works across devices. The back end will use Node.js and Express to handle things like scheduling, user authentication, and report generation. PostgreSQL will be used as the database to securely store customer information, reservations, lesson packages, test results, and activity logs.

Tools

We'll connect a payment system like Stripe or PayPal so customers can pay securely online. The system will also connect to the DMV so it can automatically update when new rules or test questions are

released. Administrators will be able to download reports in Excel format if they need to review them offline.

Infrastructure

The system will be hosted on a cloud platform like Amazon Web Services (AWS). This way, DriverPass doesn't have to manage physical servers. The cloud will handle backups, security, and scaling as the business grows. All data will be protected using secure HTTPS encryption.